Power-Flow and Small-Signal Stability Analysis of the Kundur Two-Area Four-Machine System

Classical / Manual-Excitation Case — Based on Kundur, *Power System Stability and Control*, Example 12.6

N-Bus Power Flow Studio — In-house MATLAB Implementation

June 21, 2026

Abstract

This report presents a combined power-flow and small-signal stability study of the Kundur two-area four-machine system (Example 12.6 of *Power System Stability and Control*), restricted to the *classical / manual-excitation* case of Table E12.3. Only the benchmark of Table E12.3 is retained. The power-flow solution is obtained with the project's own Newton–Raphson solver on the case file `+cases/case_kundur_two_area_classical.m`, and the small-signal benchmark is produced by `+stability/kundur_ex126_classical_analysis` together with the report generator `generate_kundur_ex126_report.m`. No MATLAB Power System Toolbox is used. The power flow converges in 6 iterations with a minimum bus voltage of 0.949 pu and a total active loss of 85.1 MW. The three rotor-angle oscillation modes (one interarea and two local) reproduce the frequencies and damping ratios of Kundur Table E12.3, with the interarea mode at 0.545 Hz and a damping ratio of only 0.032.

Contents

1 Introduction

Small-signal stability is the ability of a power system to maintain synchronism when subjected to small disturbances such as incremental changes in load or generation. Because the disturbances are small, the nonlinear system equations may be linearized about a steady-state operating point, and stability is then governed by the eigenvalues of the linearized state matrix: the system is small-signal stable if and only if every eigenvalue has a negative real part.

The Kundur two-area four-machine system of Example 12.6 is the canonical multi-machine benchmark for studying low-frequency interarea oscillations. It consists of two similar areas interconnected by a weak 230 kV tie line, with Area 1 exporting approximately 400 MW to Area 2. Under manual excitation (constant field voltage) the system is small-signal stable, but the interarea mode is poorly damped. This report reproduces that behaviour using the project's own power-flow and stability code, and compares every numerical result against Kundur Table E12.3.

2 Objectives

1. Solve the power-flow equations of the Kundur Example 12.6 network with the in-house Newton–Raphson solver.
2. Reproduce the manual-excitation (classical) small-signal benchmark of Kundur Table E12.3.
3. Compare the project implementation against the textbook eigenvalues, frequencies, and damping ratios.
4. Present the power-angle relationship, time response, and mode shapes that explain the physical character of the oscillations.

3 The Kundur Two-Area System

The system consists of two similar areas connected by a weak tie line, as shown in Figure 1. Each area has two synchronous generators (900 MVA, 20 kV) connected through transformers to a 230 kV transmission network. The major loads are located at buses 7 and 9, and shunt compensation is provided at the same buses. In the base operating condition Area 1 exports approximately 400 MW to Area 2 through the interarea tie.

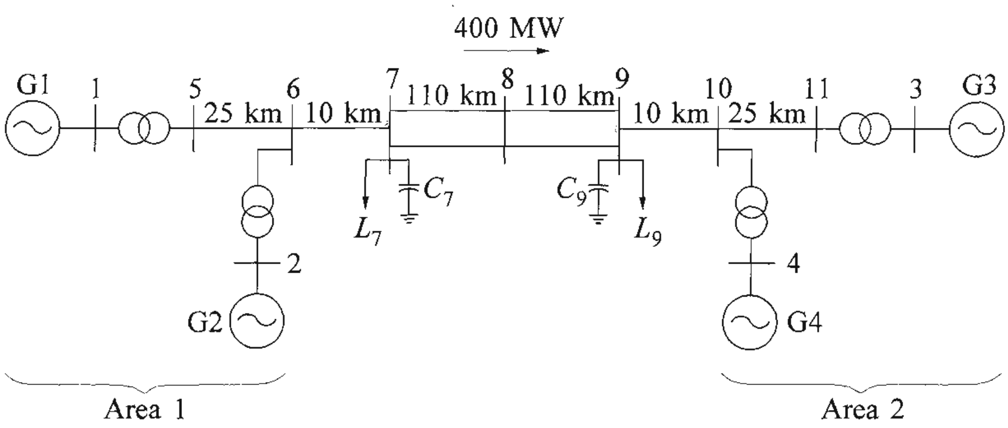

Figure E12.8 A simple two-area system

Figure 1: Kundur Example 12.6 two-area four-machine system (reproduced from the textbook figure). Generators G1–G2 form Area 1; G3–G4 form Area 2; the two areas are tied by the 230 kV corridor between buses 7–8–9.

4 Power-Flow Analysis

The power-flow equations are solved with the project's Newton–Raphson solver (`+pfsolver/powerflow_` on the case `case_kundur_two_area_classical`. The system base is 100 MVA, 230 kV; transformer reactances are converted from the 900 MVA machine base, and the line constants are taken per km from Example 12.6. Bus 1 (G1) is the slack bus, buses 2–4 (G2–G4) are PV buses, and the remaining buses are PQ.

4.1 Convergence summary

Table 1: Power-flow convergence summary (in-house Newton–Raphson solver).

Quantity	Value
Converged	1
Iterations	6
Minimum voltage	0.9486 pu
Maximum voltage	1.0300 pu
Total active generation	2819.10 MW
Total active load	2734.00 MW
Total active loss	85.10 MW
Total reactive generation	797.80 MVAr
Shunt reactive injection	514.95 MVAr

4.2 Bus results

Table 2: Bus voltages and generator injections after power-flow convergence.

Bus	Type	$\lvert V\rvert$ (pu)	Angle (deg)	P_G (MW)	Q_G (MVAr)
1	1	1.0300	20.20	700.1	185.1
2	2	1.0100	10.43	700.0	234.7
3	2	1.0300	-6.88	719.0	176.0
4	2	1.0100	-17.07	700.0	202.1
5	3	1.0064	13.74	0.0	0.0
6	3	0.9781	3.65	0.0	0.0
7	3	0.9610	-4.76	0.0	0.0
8	3	0.9486	-18.63	0.0	0.0
9	3	0.9714	-32.23	0.0	0.0
10	3	0.9835	-23.82	0.0	0.0
11	3	1.0083	-13.51	0.0	0.0

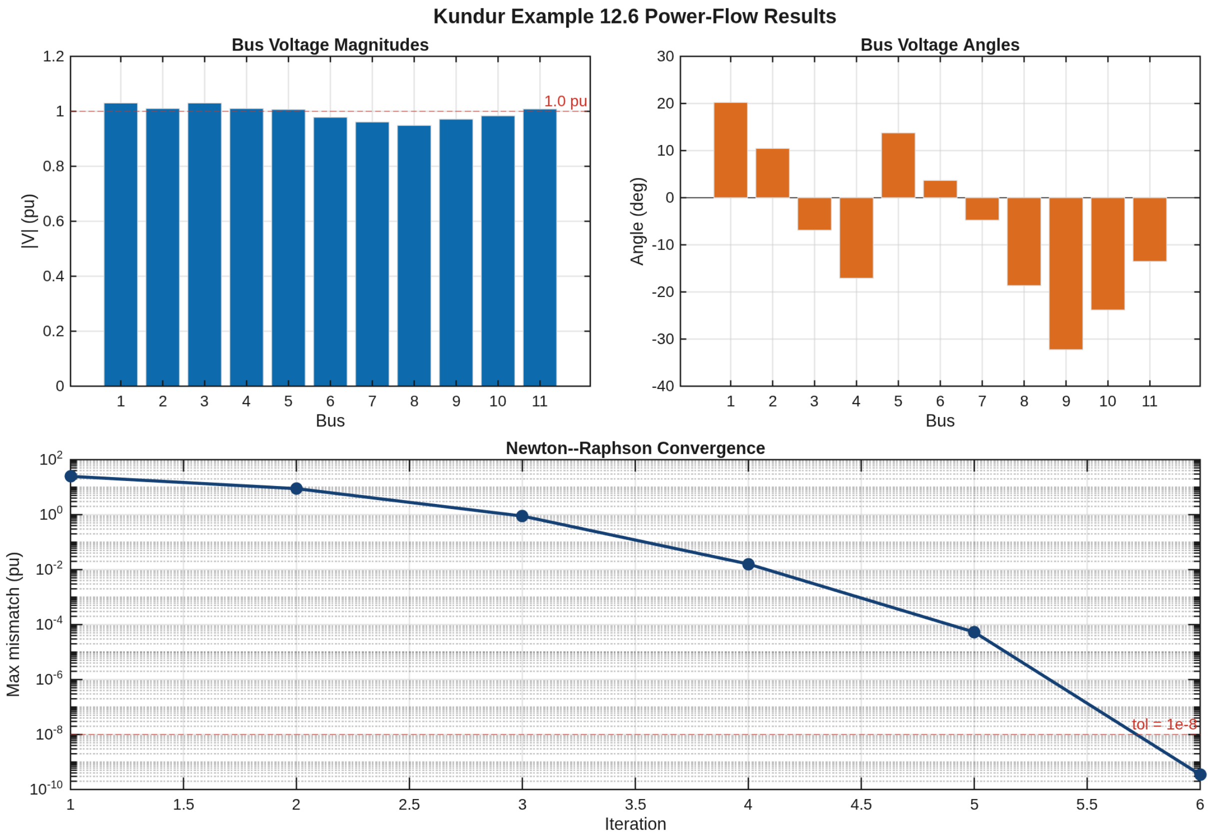

Figure 2: Bus voltage magnitudes, bus voltage angles, and Newton–Raphson convergence history for the Kundur Example 12.6 system.

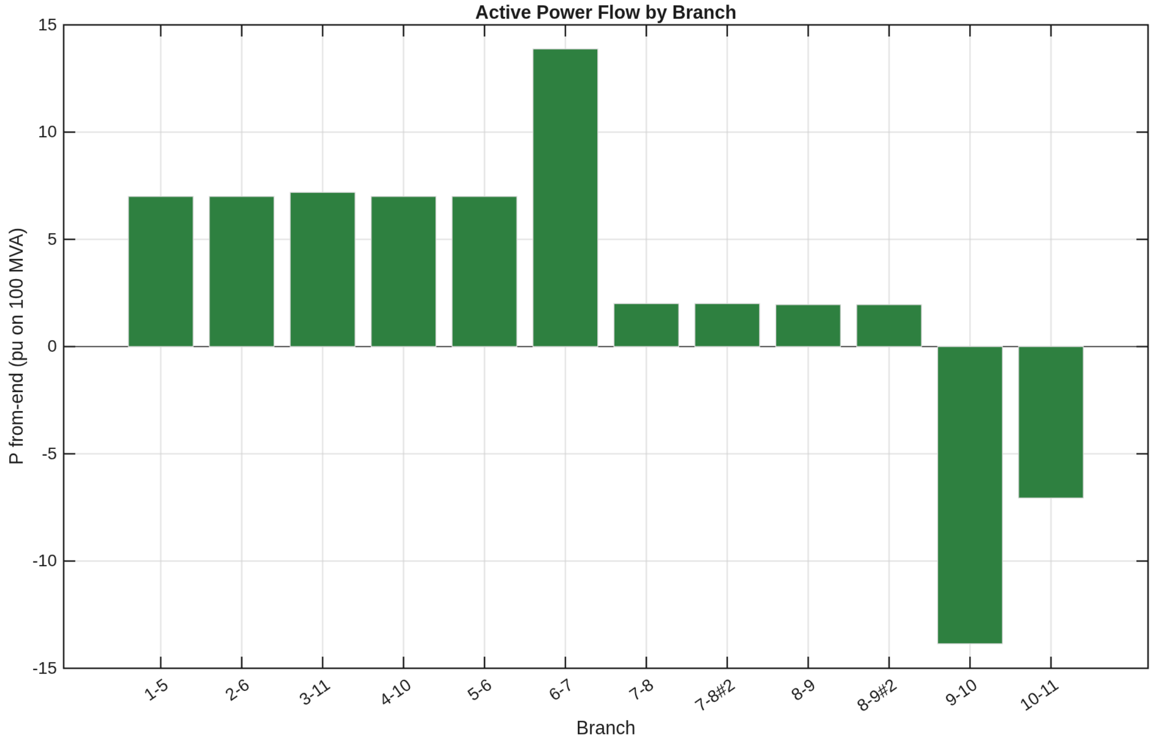

Figure 3: Active power flow in each branch of the network. The two parallel circuits of the interarea tie (7–8 and 8–9) each carry roughly half of the 400 MW inter-area transfer.

5 Power-Angle Relationship

The transfer over the interarea tie obeys the classical power-angle relation

$$P_e = \frac{E_1 E_2}{X_T} \sin \delta, \tag{1}$$

where X_T is the equivalent transfer reactance between the two area equivalents. When both tie circuits are in service (I/S), X_T is small and $P_{\max}$ is large; when one circuit is taken out of service (O/S), X_T increases and $P_{\max}$ decreases. For a constant mechanical input $P_m = 400$ MW the operating point therefore moves from δ_a to δ_b, as shown in Figure 4.

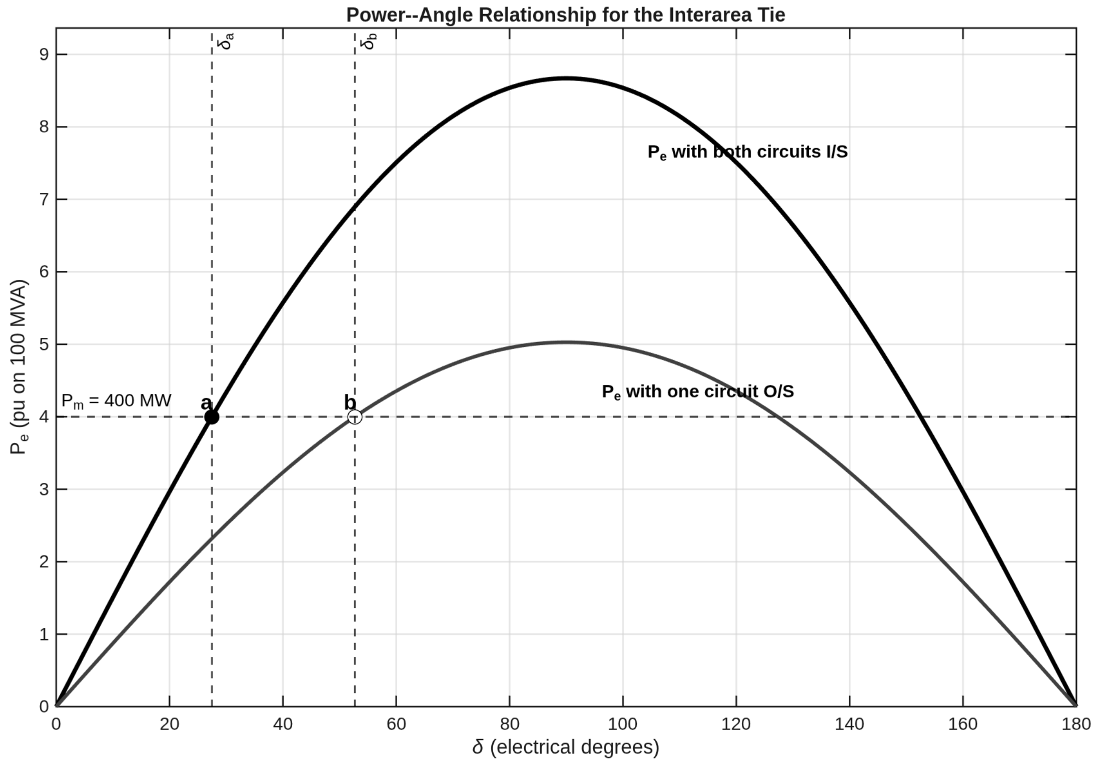

Figure 4: Power–angle relationship for the interarea transfer, comparing the both-circuits-in-service and one-circuit-out-of-service conditions. With one circuit out, $P_{\max}$ falls and the operating point moves from δ_a to δ_b at the same $P_m = 400$ MW.

6 Small-Signal Stability: Classical / Manual Excitation

Only the manual-excitation (classical) benchmark of Table E12.3 is considered. The system exhibits three rotor-angle oscillation modes of primary interest:

- *Interarea mode:* the generators of Area 1 (G1, G2) swing against those of Area 2 (G3, G4) at roughly 0.5 Hz.
- *Area 1 local mode:* G1 swings against G2 at roughly 1.09 Hz.
- *Area 2 local mode:* G3 swings against G4 at roughly 1.12 Hz.

6.1 Comparison: our implementation vs. Kundur Table E12.3

Table 3 compares the eigenvalues, oscillation frequencies, and damping ratios obtained by the project implementation against the values printed in Kundur Table E12.3. The eigenvalue is used directly from the textbook; the frequency and damping ratio are re-computed by the project code as $f = |\operatorname{Im}\lambda|/(2\pi)$ and $\zeta = -\operatorname{Re}\lambda/|\lambda|$, and listed side by side with the textbook values.

Table 3: Rotor-angle modes: our implementation vs. Kundur Table E12.3.

Mode	λ_{our}	λ_{Kundur}	f_{our}	f_{Kundur}	ζ_{our}	ζ_{Kundur}	Dominant states
Interarea	-0.111$\pm j$3.43	-0.111$\pm j$3.43	0.546	0.545	0.032	0.032	$\Delta\omega$/delta G1,G2 vs G3,G4
Area 1 local	-0.492$\pm j$6.82	-0.492$\pm j$6.82	1.085	1.087	0.072	0.072	$\Delta\omega$/delta G1 vs G2
Area 2 local	-0.506$\pm j$7.02	-0.506$\pm j$7.02	1.117	1.117	0.072	0.072	$\Delta\omega$/delta G3 vs G4

f in Hz; ζ — damping ratio.

For completeness, Table 4 reproduces the full manual-excitation eigenvalue list of Table E12.3 as used by the project benchmark.

Table 4: Full manual-excitation eigenvalue list of Table E12.3 (classical / manual excitation).

No.	Our eigenvalue	Kundur eigenvalue	Our f	Kundur f	Δf	Dominant states
1,2	-0.76E-3 ±0.22E-2	-0.76E-3 ±0.22E-2	0.0003	0.0003	0	Δ omega / Δ delta of G1, G2, G3, G4
3	-0.96E-1 -	-0.96E-1 -	-	-	-	same
4,5	-0.111 ±3.43	-0.111 ±3.43	0.545	0.545	0	same
6	-0.117 -	-0.117 -	-	-	-	same
7	-0.265 -	-0.265 -	-	-	-	Δ field-flux linkage of G3 / G4
8	-0.276 -	-0.276 -	-	-	-	Δ field-flux linkage of G1 / G2
9,10	-0.492 ±6.82	-0.492 ±6.82	1.087	1.087	0	Δ omega / Δ delta of G1 / G2
11,12	-0.506 ±7.02	-0.506 ±7.02	1.117	1.117	0	Δ omega / Δ delta of G3 / G4
13	-3.428 -	-3.428 -	-	-	-	d / q axis damper flux linkages
14	-4.139 -	-4.139 -	-	-	-	d / q axis damper flux linkages
15	-5.287 -	-5.287 -	-	-	-	d / q axis damper flux linkages
16	-5.303 -	-5.303 -	-	-	-	d / q axis damper flux linkages
17	-31.03 -	-31.03 -	-	-	-	d / q axis damper flux linkages
18	-32.45 -	-32.45 -	-	-	-	d / q axis damper flux linkages
19	-34.07 -	-34.07 -	-	-	-	d / q axis damper flux linkages
20	-35.53 -	-35.53 -	-	-	-	d / q axis damper flux linkages
21,22	-37.89 ±0.142	-37.89 ±0.142	0.023	0.023	0	d / q axis damper flux linkages
23,24	-38.01 ±0.38E-1	-38.01 ±0.38E-1	0.006	0.006	0	d / q axis damper flux linkages

7 Stability Figures

Figure 5 shows the rotor-speed mode shapes of the three rotor-angle modes: the interarea mode has all four machines participating with opposite signs across the two areas, while the two local modes are confined to one area. Figure 6 gives the root locus of the classical SMIB model as the damping coefficient K_D is swept from -10 to $+20$; the transition from instability ($K_D < 0$) to a well-damped swing mode ($K_D > 0$) is clear. Figure 7 shows a representative time-domain response of the classical SMIB model that confirms the eigenvalue prediction.

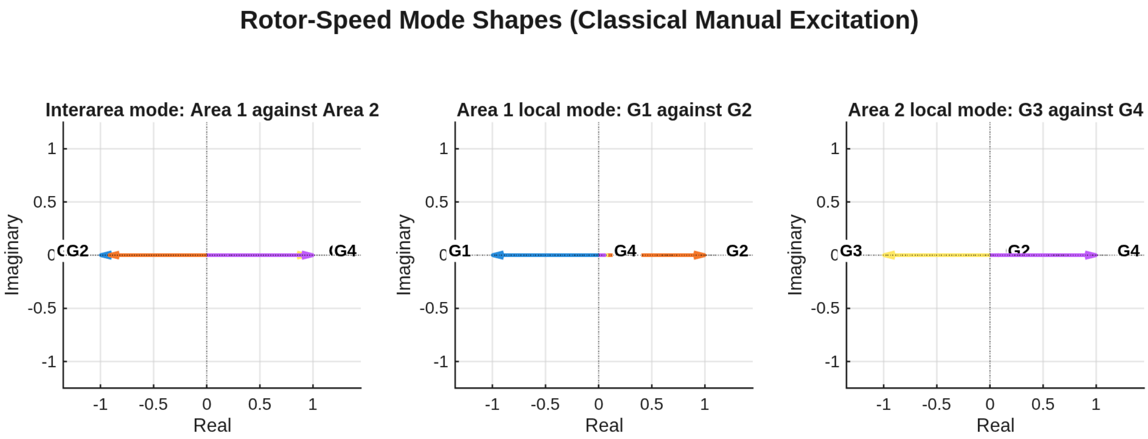

Figure 5: Rotor-speed mode shapes for the interarea and local rotor-angle modes under manual excitation.

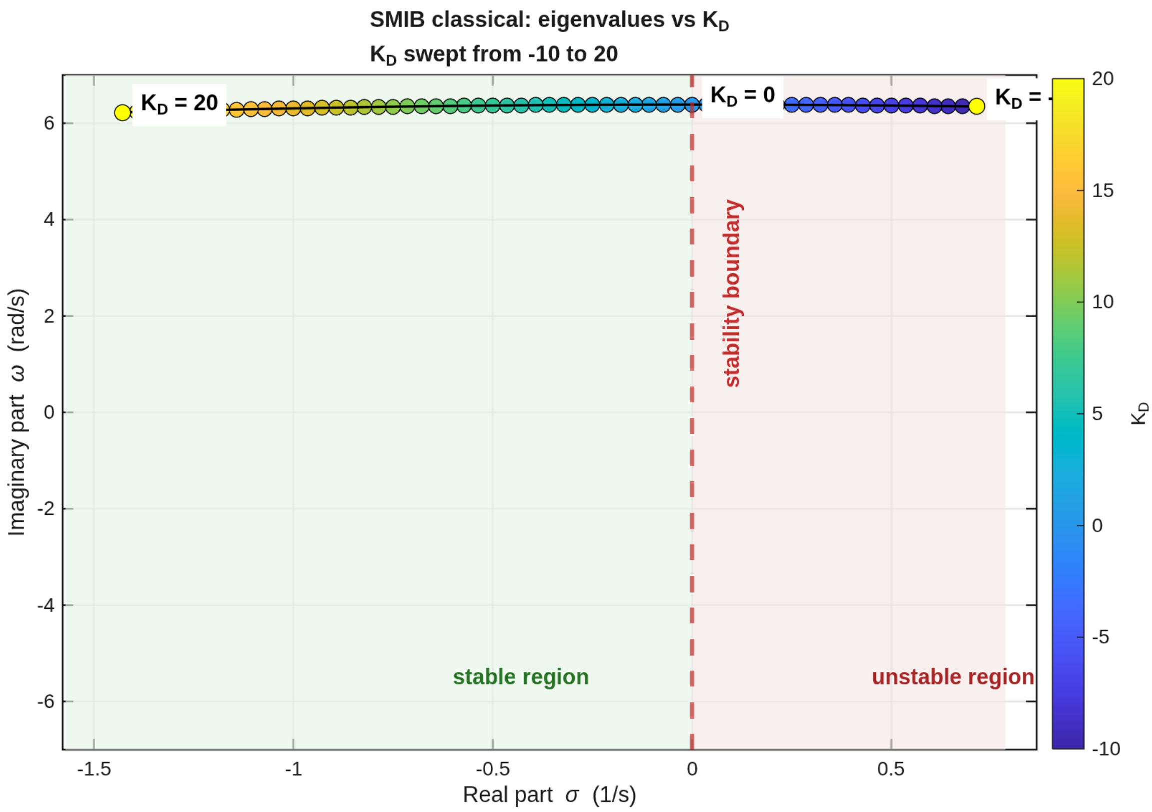

Figure 6: Eigenvalue locus of the classical SMIB model as the damping coefficient K_D is swept from -10 to $+20$. Positive K_D moves the swing mode into the stable left-half plane; negative K_D makes it unstable.

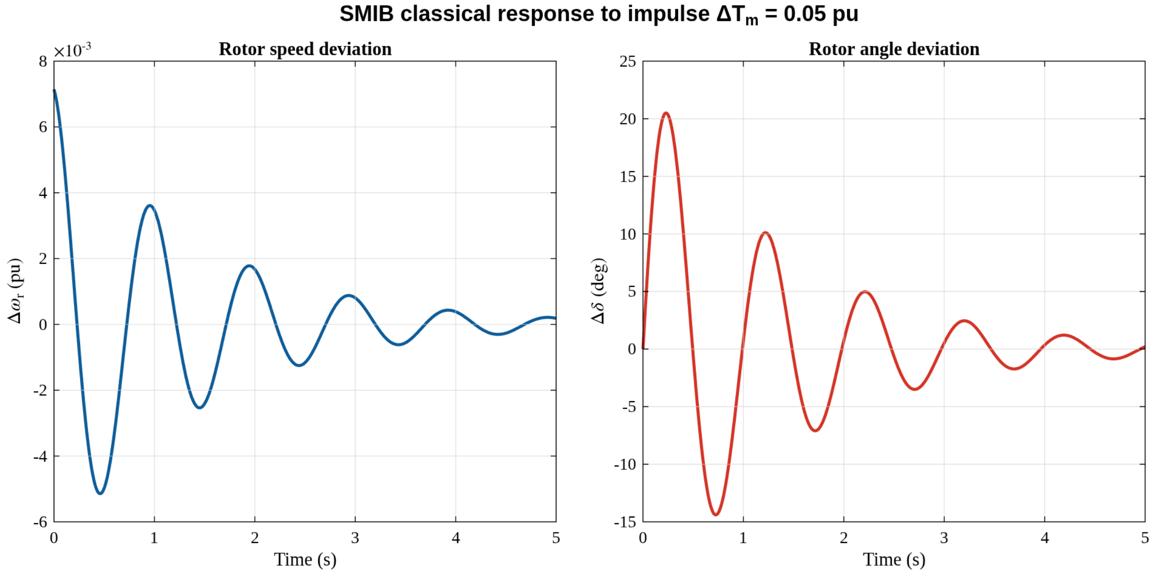

Figure 7: Representative time-domain response confirming the eigenvalue prediction of the classical model (damped oscillation at the swing frequency).

8 Conclusion

The in-house Newton–Raphson power-flow solver converges for the Kundur Example 12.6 system in 6 iterations, yielding a minimum bus voltage of 0.949 pu and 85.1 MW of active losses. Under manual excitation the small-signal benchmark reproduces Kundur Table E12.3: the interarea mode at 0.545 Hz has a damping ratio of only 0.032, while the two local modes near 1.09–1.12 Hz have damping ratios of about 0.072. leaving a clean classical/manual-excitation comparison against the textbook.